- Set the power supply to + 800 V (**be careful when increasing the bias and do it slowly).**
  - **If your detector is labeled as something other than 800 V, only set the voltage to whatever it is labeled as**
- Place a radiation source (thorium source) on a shelf below the Geiger-Mueller (G-M) tube.
- Check the amplifier output on the scope and make sure you see pulses.
  - Set the amplifier to "Neg" since the GM tube produces negative pulses.
- Set the SCA to integral mode
- Find the LLD value that discriminates all pulses
- Reduce the LLD value until you see counts on the counter/timer
  - Make sure to keep the LLD high enough so that you're only triggering on detector events (no noise triggering).

2.2 Counting Measurements

2.2.1 Measurement of a Gaussian Distribution

- With the source in place, determine a shelf height and counting interval necessary to record counts with an average value of about 30.
  - Start with a lower shelf height to ensure the detector does not have high dead time.
  - It is better to have a lower count rate and higher count time to ensure the detector is not count-rate limited.
- Record a set of 50 of these counts. You may want to use a spreadsheet program to store your data. The dwell time on the counter can be used to adjust the time between different runs. From this data, calculate:

Experimental mean:

Sample variance:

2.2.2 Measurement of a Poisson Distribution

- Use the counter to accumulate 1000 consecutive trials from a source which will give an average of about 7 counts per timing interval. The raw data don't need to be included in the lab report.
  - To get a count rate of ~7, you will most likely need to use the lowest shelf height and a low counting interval.
  - You can decrease the "Dwell" time to reduce the time between samples.
- Tabulate the data by counting the number of times 0 counts have been stored in an interval, the number of times 1 has been stored in an interval, etc. up to the maximum number of counts of any interval (make a frequency table)
- Make a frequency plot of P(i) , the probability of occurrence of exactly “i” counts vs. the value of “i” from zero to the maximum

Analysis Discussion Items

Measurement of a Gaussian distribution (Section 2.2.1)

- Include a table in the Data section showing the 50 measured values. From this, generate a histogram showing the distribution of the number of counts.
- Normalize the total area of the histogram to 1 and plot Gaussian and Poisson distributions on the same plot. For the Gaussian distribution, you assume that the mean is your experimental mean value and the standard deviation is your experimental standard deviation. For the Poisson distribution, you assume that the mean is your experimental mean and the variance is the same as the experimental mean.
- Apply the Chi-squared test to test the departure of the experimental data to Poisson distribution (follow the procedure in the textbook from page 80 to 82):
  - Calculate the probability that a random sample from a Poisson distribution with the same mean will show a larger fluctuation than that of the measured data (hint: use figure 3.11).
  - Is the Chi-squared value what you would expect? Why?
- Discuss how well your measured data fits the Gaussian and the Poisson distribution.

Measurement of a Poisson distribution (Section 2.2.2)

- Include a histogram showing the distribution of the 1000 data points (x-axis shows number of counts, y-axis shows frequency)
- On the same plot, overlay a Gaussian and a Poisson distribution (same as the analysis for Section 2.2.1)
- Apply the Chi-squared test to test the departure of the experimental data to Poisson distribution (same as the analysis for Section 2.2.1)
- Discuss how well your measured data fits the Gaussian and the Poisson distribution.

Lab Notebook Figures

Include a copy of the Gaussian and Poisson frequency distribution with fits in your lab notebook.